

WHAT IS FRUGAL, WHAT IS INNOVATION? *TOWARDS A THEORY OF FRUGAL INNOVATION*

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ABSTRACT

Innovation in emerging markets offers fertile ground for theory development. In recognition of the growing trend in “frugal innovation” discourse among practitioners particularly in emerging economies, we parse “frugal innovation” into "frugal" and "innovation" separately and present the underlying meanings towards understanding "frugal innovation" in historical and contemporary contexts. We further develop a theoretical model of frugal innovation by applying existing theories to emerging market contexts. We do so by merging technology innovation, institutional innovation, and social innovation literatures to argue that frugal innovation lies at the intersections of these streams. We show how using the model to understand what the space currently looks like may help to offer a consolidated and encompassing theory of frugal innovation and aid in opening a research agenda.

Keywords:

Frugal innovation, emerging market, resource constraints, affordability, institutional voids, theory.

Word count: 11,150

INTRODUCTION

While the economic growth of emerging markets draws MNCs from developed countries to target luxury goods to the richer segments in emerging markets, others are seeking to have greater impact by serving the 'bottom of the pyramid' and lower middle class segments.

Despite the growth of high income classes in the large emerging BRIC economies, the new generation of goods being targeted at and sold to them, will not generate enough earnings to fund both the shift towards mass production, and the development of the next technology generation of consumer goods (Soete, 2008). The reality in India for instance, is that it is not uncommon to see a family of five riding on a single Vespa scooter. While MNCs like Mercedes and BMW find inroads to target the luxury market, Ratan Tata, entrepreneur and CEO of TATA conglomerate looked at this unsafe, yet very common practice of transportation and commented:

"I observed families riding on two-wheelers - the father driving the scooter, his young kid standing in front of him, his wife seated behind him holding a little baby...It led me to wonder whether one could conceive of a safe, affordable, all-weather form of transport for such a family. Tata Motors' engineers and designers gave their all for about four years to realise this goal. Today, we indeed have a People's Car, which is affordable and yet built to meet safety requirements and emission norms, to be fuel efficient and low on emissions." (Foster and Malhotra, 2008: online)

The result, a \$2K Tata Nano car targeted to those on scooters who can now make what was once considered a dreamy leap to car ownership. The car is currently sold in India and neighboring Sri Lanka. A European version built on the Nano platform called the Pixel was showcased in Coventry, UK with innovations such as all four wheels that turn ninety degrees to park in tight spots of the UK. The market for small affordable cars has emerged.

The non-profit One Laptop per Child programme seeks to develop laptops at \$100 to give away to the 2 billion school children of the world. The project has, arguably, been

gradually supplanted by the emergence of the netbook computers prompted by Intel's announcement to contribute to the cause, albeit through for-profit models. While the One Laptop per Child helped create the market for netbooks, it has been stuck at levels of \$200 per laptop versus the target price of \$100. In contrast, several netbooks in the US are now sold at prices as low as \$150. Despite its shortcomings, the OLPC catalysed the market for netbooks, and arguably, even tablet PCs.

These two and several other examples have triggered in the news media and practitioner literature much talk on the potentially disruptive and transformational role of “frugal innovation” (Woolridge, 2010) not only for emerging markets but also developed markets (Immelt, Govindarajan, and Trimble, 2010). We first outline the increasing trends in innovation in emerging markets as fertile ground for theory development. Then we attempt to parse “frugal innovation”, and try to understand "frugal" and "innovation" each separately before merging the two towards understanding "frugal innovation". We provide historical examples from the 1940s in Britain to show that although frugal innovation motives and activities are not entirely new, today's contextual circumstances incite renewed interest and use such resourceful practices. Finally we develop a theoretical model, based on applying existing theories to emerging market contexts, to position and understand what it means to frugally innovate.

INNOVATION IN EMERGING MARKETS

Previously the wealthy countries were considered focal markets for the world, but now the double edged phenomenon of economic growth in emerging markets combined with recession and slow growth in wealthy nations is forcing much attention to be redirected to populace markets in emerging markets. Compare the consistent 8-9% growth of BRIC nations with the slow 1-3% growth of developed nations. On 27 Dec, 2011, as of writing this paper, the Brazilian economy overtook the UK to become the sixth largest economy in the

world. Some among the business community are concerned that too much inequality is limiting the sustainability of long-term growth. The demand of high income groups is, in absolute terms, increasing at a much slower rate than that of lower income groups and this is especially true for the BRICS (Fu, Soete and Sonne, 2010).

Porter and Stern (2001) cited Singapore, Taiwan, South Korea, Israel, and Ireland as the new centers of innovative capacity outside of the OECD countries. Porter and Stern denied the innovative capacity of emerging nations. They claimed:

“Conversely, several countries that have drawn much attention as potential economic powers — India, China and Malaysia — are not yet generating meaningful levels of world-class innovative output on an absolute or relative basis. These countries have developed neither a base for innovation nor clusters with a large innovative capacity.” (Porter and Stern, 2001: 34).

They further claim that although a location may be favorable for other reasons which are non-innovative capacity, such as offering low manufacturing costs or access to key markets, but would be unfavourable for innovation. Given the theoretical lens of the Diamond of Competitive Advantage Porter and Stern employed, their argument may be understandable. As such, it is often proclaimed that the developed nations maintain a distant edge in innovation capabilities as compared to the emerging nations.

But the importance of innovation among the latter is prevailing. In the 2010 Bloomberg BusinessWeek’s 50 Most Innovative Companies rankings, for the first time since the annual rankings were launched in 2005, the majority of corporations are based outside the U.S and the new global leaders are coming out of mainly Asia. In terms of executive leader mind-set in China, 95% of executives said innovation was the key to economic growth, while 90% and 89% of respondents in South America and India, respectively, agreed. Comparatively lower, in the U.S., only 72% said innovation was important (Arndt and Einhorn, 2010).

We suggest this contrary to widely held belief of growth in innovation in emerging markets is better looked at from an alternative theoretical lens. It may be that emerging nations are approaching innovation in a different way that addresses contextual factors, constraints and local demands. So an alternate theoretical lens and accompanying evidence may prove to the contrary. Popular innovations that meet local emerging market demands include the TATA NANO car that costs less than 3000 dollars, GE's mini-handheld ECG machine in its Bangalore R&D centre that costs less than half of conventional bulky ECG machines, and Bahria Town's 5000 dollar homes in Pakistan. Tata stands at rank 17 on BusinessWeek's most innovative companies list while GE's local division in India ranked 9 (Bloomberg BusinessWeek, 2010).

One side says not enough basic R&D is done in emerging markets (Porter and Stern, 2001) which is relatively true as compared to the wealthier nations. But the flip side is that a different type of innovation is taking place, one that embodies 'frugal innovation' activity which attempts to serve large bottom and lower middle class population demands. And this is contrasted with the top-down sophisticated R&D led innovation to one that employs bottoms up, human centric, appropriate, local, and cost efficient approaches through processes such as design thinking, bricolage, creative improvisation, lean and reverse engineering. Although none of these concepts are independently new, but the shift in all working together through varied actors is what is solving the underserved needs and helping to build capacity for companies and nations.

In the basic sense, frugal innovation has always occurred since the invention of Neanderthal hand tools from stones and bones to making do with what is on hand. Innovation in its most basic form is an old practice that has permeated our human make-up. However it is gaining renewed attention given economic, resource, and demographic shifts. Much of

today's challenges in developing countries are chronic and may have historical precedence, indeed even in the West as it sought to grow and develop.

UNPRECEDENTED? FASHION IN 1940S BRITAIN

The circumstances we see today that help identify frugal innovation are not entirely unique. British society in 1940s was marked by austerity measures to assure necessary resources were committed to winning the war. But rationing meant diminished supply for the civilian population, many of whom affected were women and children. The government worried about escalating consumer prices which risked not only societal inequality but also low public morale needed for winning the war.

The Civilian Clothing 1941 (CC41) utility scheme introduced by the British Board of Trade in 1941 sought to ensure quality consumer goods were available at reasonable prices. It was first introduced for clothes and later to furniture but the frugal principles extended to any consumer items. The civilian scheme forbade use of excessive material and of certain chemicals and materials like wool which was needed for military uniforms. Everything had to last as long as possible and being wasteful was considered detrimental to winning the war. The scheme sought to bolster a "make do and mend" frugal mentality which transpired in what was deemed "good" design and style, high efficiency and quality, but still affordable and accessible to all British segments.

In clothing, the CC41 austerity regulations introduced dress restrictions, such as no more than 2 pockets, 5 buttons, 6 seams in a skirt, 2 inverted or box pleats or 4 knife pleats and no more than 4 metres of stitching (Hull, 2011). However there were no fashion restrictions. Given the lukewarm public reception to the rationing and restrictive scheme, the government commissioned some of London's top fashion designers to create year round collections. The Incorporated Society of London Fashion Designers created 34 smart Utility Clothing designs in 1942 (Thomas, 2011).

The CC41 scheme was complemented by encouraging a "make do and mend" culture propagated by "Mrs Sew and Sew", the handy seamstress character. Mrs Sew and Sew encouraged women to be creative with their dressmaking skills. Films by the government showed how to make old items into new ones such as a baby's cot from old sacking, dressing gowns from patchwork, and ladies' dresses from men's dress clothes (Rationing, 2011). Women improvised by turning drapes and bedding into clothes, pillowcases into white shorts for summer, and wedding dresses into French Knickers or nightgowns. Sometimes alternative materials were used and sometimes none at all by introducing novel substitute practices. Given rubber and leather was needed by the war effort, shoes were made with soles made of cork. And with nylon and silk needed to make parachutes, women improvised by painting their bare legs to simulate stockings (Rationing, 2011).

But improvisation was not limited to use of materials. Many designers sought to redesign dresses by inventing new styles while adhering to the CC41 restrictions. Pleats, cuffs, ribbons, and decorations were incorporated in ways that were not outright forbidden. As one blogger says about the inventive fashions of war-time, "I love it -- the removal of anything extraneous, but the attempts to still add detail" (Mags, 2011: para 1).

Since chemicals were in short supply, make up was unavailable, yet women felt the need to look beautiful as a morale booster. Jodie Kidd, the fashion model turned TV anchor narrates in the television show "Ration Book Britain", women in the 1940s "put beetroot on their faces to make them look rosier and healthier. They painted Bisto on their legs because they couldn't get stockings and used a burnt match as eye shadow and eyeliner" (Coventry, 2010, para 4). Other tricks of improvisation included using cherry juice for blusher, beetroot for lipstick, and wiping used teabags over skin to change a white complexion to a bronzed look (Ration Book Britian, 2010).

Notwithstanding the rationing of timber, the government wanted the quality of furniture to be high, yet affordable and accessible to the majority of British consumers, particularly the newly-weds. Furniture design was standardized, mass produced in modular fashion, and delivered half completed to consumers with do-it-yourself (DIY) assembly directions. The CC41 affordable yet durable DIY furniture continued in popularity even when the scheme was lifted in 1952 (Ration Book Britain, 2010). This IKEA type furniture can be considered a "frugal innovation" with benefits that extended far beyond the circumstances that triggered it.

Fashion historians argue that although the utility clothing line was produced under strict regulations and first met with scepticism, the new fashion did not sacrifice style but rather redefined it (Schenk, 1994; Mendes, 1999; Mendes and De La Haye, 1999; Ration Book Britain, 2010). The utility collection design was praised in British Vogue in 1942 for "the clean elegance of a style stripped of all superfluities" (Mendes, 1999: 45). The efforts towards "the removal of anything extraneous, but the attempts to still add detail" (Mags, 2011: para 1) rings similar to frugal innovation proponents. Much of this redesign of clothes and furniture meant co-opting some of the times most popular fashion designers. But while popular fashion designers, albeit even today, design for small exclusive batch production, the government sought to promote high quality design to the masses. Dover (1991) credits the Design Research Unit setup in 1942 soon after the utility scheme in 1941 for introducing the term "industrial design" into general usage by championing for its value towards the economy and winning the war. The subsequent establishment of the Council of Industrial Design by the Board of Trade in 1944 helped professionalize the practice by educating the public about good design, product endorsements, and exhibitions exhorting "Buy wisely in Britain" (Design Council, 2011).

We see that Britain in 1940s brought about new institutional policies, new technological and design techniques, and new social consumer practices. Kidd posits in the television series that the spirit of winning involved making the most of what you had, that the 1940s showed how austerity can lead to innovation and a greater sense of resourcefulness. The upshot of this problem-turned-opportunity is explained by the need to (i) conserve raw materials such as cloth, wool, leather, wood, chemicals, etc, (ii) make manufacturing more efficient since much of the skilled labour had left to fight, and (iii) increasing prices needed to be controlled for the civilian population to afford good at good quality (Bayley, 2011).

All societies no matter how resource constrained, try to add value by solving their problems or to meet needs. What is new today is the extent to which the practice is being understood as a means to solve long withstanding problems and fulfil unmet needs. The mix of challenges and opportunities that emerging market contexts pose offer new possibilities to recombine different resources to make frugal innovation happen in a better way with greater impact. Working around lack of resources, skills, and the ability to still meet demand with affordable high quality solutions are what many of today's "frugal" innovators seek to accomplish.

THE NOTION OF CONSTRAINTS IN EMERGING MARKETS

We believe that emerging markets offer a unique perspective on understanding how innovation itself evolves. These environments offer a unique context for innovating given the contextual constraints of functioning within. Three main challenges persist in emerging nations for innovation: First are resource constraints, second is the challenge and opportunity of dealing with institutional voids, and third is the need to address the needs of the bottom of the pyramid i.e. the largest and poorest socio-economic segment of the population (Prahalad, 2005). Despite institutional voids, emerging market entrepreneurs and firms are producing innovations which are resolving their local needs, and at the same time profiting to the extent

that they can expand to neighboring developing nations and even beyond to developed markets (Khanna & Palepu, 2006).

We call these environments extreme as ventures operating therein seek to mitigate or adapt often simultaneously to affordability, resource, and institutional constraints. An environment is resource constrained if it provides new challenges, whether opportunities or problems without providing additional or new resources (Baker and Nelson, 2005). Stiglitz (2001) posits that all societies are resource-constrained and poor countries even more so. The procurement, control and combination of labor, skills, and material is crucial to the creation of new products and services (Schumpeter, 1934; Shane & Venkatraman, 2000). Yet emerging and developing markets can be considered extreme environments given the penurious nature of basic facilities such as infrastructure, literacy, access to literacy, medical care, retail chains, communication networks, transportation, housing, and sanitation.

In moving beyond resource constraints, Stiglitz (2001) suggests that on top of the general resource constraints faced by developing countries are the constraints on the capacity of government to deal with the number of issues it can pursue. That limitation to cope with issues poses both an institutional challenge and opportunity. Institutions can be defined as the humanly derived constraints that structure how humans interact. These constraints can be formal constraints as in the case of formal rules, laws, and constitutions, or they can be informal constraints as in the case of norms of behaviour, conventions, and codes of conduct (North, 1997). Institutional concerns such as legal recourse and political structure are global concerns for anyone instituting change and innovation and perhaps more so in emerging markets given the institutionally complex contexts (Mair, Marti and Ventresca, 2011). Yet, perhaps for these challenges, these are the environments where social enterprises look to provide solutions for and operate in (Mair & Marti, 2009; Sarasvathy, 2006).

A third constraint or challenge and opportunity in emerging markets has to do with the social dynamics of vast number of populations living close to poverty. One billion people live in the least developed countries and four billion live in developing countries (Collier, 2007). This large segment poses a challenge for multinational corporations (Prahalad, 2005), entrepreneurs (London and Hart, 2004), and governments (Sachs, 2006) alike to provide affordable solutions that help mitigate poverty and its consequences. All three sectors need to and are in many ways joining hands to innovate for the global low-income population.

Although, the BOP has been a tacit market argument that projects consumers as waiting out there to be served, nevertheless we believe the BOP literature has provided an anchor to talk about frugal innovation. Rural markets and especially the BOP markets can be hotbeds of innovation (Hart and Christenson, 2002; Prahalad, 2005). The BOP market can test a company's capabilities since it is comparatively difficult to market to the BOP than their rich counterparts not only in scale of operations but also in scope and sustainability (Tripathi and De, 2007). Feedback from BoP users and from design developers upstream might result in reverse transfer of technology (from the South to the North), re-invigorating and motivating the research community in the highly developed world increasingly "in search of relevance" (Soete, 2008). President of PATH, an international non-profit working for health solutions, writes in the Lancet:

"We often assume that these frontiers of science will benefit only the richer nations of the world, ...[But] in fact resource-poor settings can actually drive innovation, demanding ingenious product designs that are less expensive, and easier to use, and require less infrastructure. It is also easier to disrupt the technological status quo in the absence of entrenched commercial interests organised around existing products" (Elias, 2006: 540).

The technologies and capabilities developed for the bottom may have the opportunity to 'move up the pyramid' and challenge existing capabilities too often developed first for the top-of-the-pyramid markets thereby offering the missing means by which firms discover

creative destruction (London and Hart, 2004). London and Hart further believe that it is possible for reinventions that seek to meet the requirements of the base of the pyramid to eventually help solve some of the developed nations' most pressing environmental and social problems. Existing business models that reflect the over-consumptive nature of existing top-of-the-pyramid strategies (Hart, 1997) could not conceivably fulfil the needs of 4-5 billion at the bottom without increasing the rate of depletion of limited global resources.

Furthermore, some argue that the emerging markets offer a unique opportunity to gain competitive advantage (Khanna and Palepu, 2006). *“Since modern firms must be capable of dynamic responses to innovation, consumer tastes and competitive challenges (see Anderson, 1995; Schoenberger, 1997, 2000; Monk, 2008), this (emerging) market will provide its inhabitants with a potential advantage.”* (Monk, 2008, p.2) The outcome may be that frugal innovation as a strategy can provide an edge for adopters.

For instance, GE is trying to disrupt itself by departing from a glocalization strategy to one that of reverse innovation which stems from emerging markets (Immelt et al, 2009). By shifting from a strategy where product comes first and country second and working the other way around, GE is trying to create new markets both in developing countries and for new applications at home (Immelt et al, 2009). Old business models can serve as constraint for innovation while new constraints, as those faced in emerging markets, can help trigger new business models.

For EM contexts, entrepreneurs and innovators have to devise low cost strategies to deal with or circumvent institutional voids and resource limitations to innovate, develop and deliver products and services to low income users with little purchasing power, often at mass scale. We use the term extreme environments to denote these contexts where any or all three contextual constraints exist i.e. affordability, resource, and institutional constraints. Emerging and developing markets face all three which makes them an ideal context to study these new

trends in innovation. Lack of essential pieces of the ecosystem may serve as constraints thereby making it difficult to design and deploy new products and services.

Although emerging market conditions mobilize us to look through the lens of resource constraints, institutional voids, and affordability constraints, these conditions are not unique to emerging or developing markets. There may exist pockets of regions or market segments within developed countries which also hold the same challenges or opportunities. Consider for instance the native American or native Australian tribes whom by account of their unique status in their respective federal systems make them economically and institutionally distinct.

So we shall see there are examples of innovations that have had to deal with any of the constraints or a combination thereof, in both developing and developed world markets even though our theoretical lens begins from prevailing circumstances in emerging economies. But before we propose a theoretical model for understanding “frugal innovation”, let us attempt to parse the term into “frugal” and “innovation”.

WHAT IS FRUGAL?

Frugal in the literal sense means sparing, thrifty, or "characterized by or reflecting economy in the use of resources" (Merriam Webster, 2011) and "simple and plain and costing little" (Oxford Dictionaries, 2011).

Lastovicka et al (1999) seminally reviewed and studied frugality and presented a conceptual definition as well as a measurement scale. Their literature review of mainly consumer research and marketing literature revealed neglect of the phenomenon, and when they broadened the search to all the social sciences and humanities, including business-related journals, still no significant academic literature was revealed. They thus covered frugality from the religious, economic, self-help, and the psychological perspectives. They find frugality is entrenched in the human past. It was widely prevalent in early America

(Lastovicka et al, 1999). The Boston Evening Post of 1767 exemplifies this practice by urging "Save Your Money and You Will Save Your Country!" (Morgan, 1967).

Most religions promote ascetism or restraint from materialist desires. However, frugality is presented not as deprivation but rather as "sacrificing a series of whims for the sake of obtaining a more worthy goal" (Lastovicka et al, 1999: 1987). According to Nash (2000: 169; Pepper et al, 2009) frugality is an 'ethically conscious' choice, 'intentionally responsive to social and ecological conditions' of 'excessive and unfair consumption and production'. People wishing to adopt frugal practices such as say recycling, can only do so effectively with institutional support from the wider society. Recycling needs public acceptance, recyclable products, recycling centers, market potential and supportive public policy (Nash, 2000). So for both individuals and societies, frugality as an economic activity favors ethical and disciplined action "for the sake of some higher ends" (Nash, 2000: 173).

From the economic perspective, Adam Smith (1904; II.3.19) held that capital originates in savings by the frugal: "*By what a frugal man annually saves, he not only affords maintenance to an additional number of productive hands, for that or the ensuing year, but, like the founder of a public workhouse, he establishes as it were a perpetual fund for the maintenance of an equal number in all times to come.*" Private frugality can help in growth of capital and in the eventual increase of national opulence. John Stuart Mill (1848) argued for a theory of capital based on frugality. Mill believed the frugal few look to a greater future return as a reward for current abstinence and are therefore the source of capital. The psychological perspective and the self-help literature reveal clever and resourceful use and reuse of products and services as a characteristic of being frugal. DeYoung on appropriate use of natural resources (1986: 285) defines frugality as "careful use of resources and avoidance of waste." So frugality is not just about what is acquired, but also how something is used by embracing reduction or elimination of waste. From the behavioral science

perspective, Lastovicka et al defined frugality as the "degree to which consumers are both restrained in acquiring and in resourcefully using economic goods and services to achieve longer-term goals" (p.88).

Unfortunately perhaps, the label frugal often conjures up images of Ebenezer Scrooge, the selfish and miserly protagonist of Charles Dickens' 1843 novel "A Christmas Carol". However, Lastovicka's et al (1999) research shows frugality is not pure deprivation for the sake of hoarding. Rather it reflects short-term pragmatic sacrifices to achieving longer-term goals. The reality is that whether by religious orientation, philosophy, or necessity, frugality is practised by a vast majority of consumers in the developing world and even in the developed world. Firms adopt frugality at times of reduced recessionary revenues or competitiveness induced squeezed profits. So are governments being called across Europe and even the US to reduce spending and increase focus on activities for long-term return.

Simply put, a frugal philosophy embodies "doing more with less". This applies at both the level of the buyer and seller. From the consumer perspective, a frugal solution is low cost, affordable. But affordability extends beyond simply the cost of the solution to operational and disposal costs. A product or service needs complementary solutions, resources and infrastructure to continue performing to its worth. Frugality means not just lowering the cost of the product, but also in how it is designed to operate in the resource constrained context in which it functions.

Thus for the consumer a frugal solution extends from simply cost to functioning with few resources, against lack of necessary infrastructure, and in how it works within complex, different, or lackluster institutions. From the firm perspective, a frugal solution has to be designed, produced, delivered, and maintained to achieve the needs of underserved consumers in constrained environments. The firm itself may have to function in an environment marked by one or all of constrained resources and institutions.

WHAT IS INNOVATION?

Innovation is both an outcome (product) and process as academic literature is split into two broad streams. The outcome stream manifests innovation in new products, product features, and production methods. Research in this stream studies the sources and economic consequences of innovation (von Hippel, 1988; Abernathy and Utterback, 1978). The process stream studies social and organizational processes that produce the outcome innovation, such as individual creativity, organizational structure, environmental context, and social and economic factors (Phills, Deiglmeier, and Miller, 2008; Kanter, 1984; Amabile, 1988).

Even Schumpeter wavered in his support for entrepreneurs versus large corporations as a means to the end of innovation, yet it was innovation that triggered waves of "creative destruction". Furthermore, innovation, unlike entrepreneurship or the enterprise, helps to transcend levels of analysis, methods, disciplines, and sectors to reveal processes that produce solutions to problems. Hence innovation can stem from people and places beyond just the entrepreneur or enterprise, through for instance start-ups, non-profits, large firms, and even governments.

As Phills, Deiglmeier, and Miller (2008) argue for social innovation, we focus on the innovation itself, rather than the entrepreneur or organization. Although Yunus was the entrepreneur who pioneered microfinance and Grameen the organization through which it was scaled, microfinance as an innovative solution is what solved a social problem and created value for society as a whole. By focusing on the innovation, rather than the entrepreneur or organization, we can gain a better understanding of the mechanisms that result in value creation (Phills et al, 2008).

Two criteria help identify innovations, whether process or outcome based (Phills et al, 2008). One is that the phenomenon must be novel, though not necessarily purely original, it must be new to the user, use, application, or context. The "novelness" associated with

innovation does not have to be universal, but can be new to a specific firm or market. March and Simon (1958) argued that innovations often transpire by borrowing from other innovations rather than through rudimentary invention. Van de Ven (1986) views innovation as a new idea or recombination of old ideas that to the people involved are perceived as new, even if to others it seems to be an imitation of something that exists elsewhere. Second is that the innovation should entail improvement by being either more effective or more efficient over solutions sought to be replaced. Innovation involves sophisticated and often capital intensive research and development (R&D) by industrial organizations but also incremental improvements to existing technology by just about anyone. The “who”, “what”, and “how” one innovates is varied from firms to users, products to new business models, and basic research to acquisition of external knowledge. In this sense, innovation does not necessarily have to be entirely new or to need huge financial and human capital requirements often associated with high-tech R&D. As a third component, we add the notion of frugal, i.e. doing more with less.

Frugal innovation can encompass both processes and outcomes and thereby has overlapping meanings. It can refer to frugal processes of innovation, such as the process of reverse diffusion (Govindarajan and Ramamurti, 2011), reverse engineering, use of bricolage (Levi-Strauss, 1967), creative improvisation or *jugaad* (Gulati, 2010), design thinking processes, and use of tools such as open source techniques. Jugaad is presented as an improvisational style of innovation driven by scarce resources and attention to the immediate needs of customers rather than their lifestyle wants (BusinessWeek, 2009). Cisco's chief globalization officer Wim Elfrink, moved from San Jose to Bangalore in 2007 and says, "The innovation agenda in India is affordability and scale" (BusinessWeek, 2009: online).

But while frugal innovation and *jugaad* both can refer to innovating under for instance cost or institutional constraints, frugal innovation instead endeavours to maintain or

surpass performance dimensions. The needs of underserved low purchasing power consumers when combined with challenges in extreme environments means expectations of performance can be high but also socially beneficial. As such we deviate from definitions that focus solely on low-cost. It is not simply about reducing cost, but can also involve increasing the affordability power of the buyer through income generation, saving, or alternative payment schemes. Frugal innovation may also mean that the outcome involves building local entrepreneurship, capacity building and self-reliance or sustainability.

In essence frugal innovation is a label that captures a range of heterogeneous activities which cut across different sectors (see our other working paper on Frugal Innovation: Fad, Fashion, or Fit?, 2011). For a product, service, practice or process to be considered as frugal innovation, conditions called for by each of the two components need to be met. The frugal aspect involves solving needs without being stymied by affordability, resource, or institutional constraints. The innovation aspect involves innovating at one of the intersections of technological, institutional and social innovation.

We propose a definition of “frugal innovation” that beyond costs, more holistically looks to create value for underserved markets. As such frugal innovation may redefine business models, reconfigure value chains and redesign products to use resources in different ways and create more inclusive markets by serving users with affordability constraints, often in a scalable & sustainable manner.

THEORY DEVELOPMENT IN EMERGING MARKET CONTEXTS

The contention that there is little known about the growth of the innovation phenomenon in emerging markets as innovation research has largely focused on mature markets (Drazin and Schoonhoven 1996) may not hold true anymore. A search on “innovation” and “emerging markets” or “developing markets” in EBSCO Business Source reveals 294 results. Yet academic literature has still not tapped the potential of understanding innovation in emerging

markets by applying and testing theories relevant to contextual factors important in those markets. Drazin and Schoonhoven had further challenged researchers and students of innovation to not just perform studies on innovation in emerging economies but to test them to those environments to counter the charge that theories of innovation are ethnocentric and uninformed by internationally based comparative research.

Domains such as strategy have shown that researchers should not assume that theories or findings in a developed economy will be equally of relevance in an emerging economy (Peng and Lou, 2000). It is therefore, argued that theory may need to be adapted in new settings (Young, Peng, Ahlstrom, & Bruton, 2002). Shaker Zahra (2007) posits that if the contextual nature of emerging economies is considered then insights on theory can be generated to better understand emerging economy entrepreneurship (and innovation) or even generate new theory.

Few authors have explicitly looked at whether entrepreneurship in environments of extreme scarcity is different from conventional business entrepreneurship (Desa, 2009). We highlight that even fewer authors have looked at whether innovation in environments of extreme scarcity marked by low affordability, inaccessibility, limited resources, and lack of ‘mainstream’ organized institutions is any different from conventional business innovation.

Prahalad and Mashelkar (2010) argue that most innovation programs are built on the assumptions of affluence and abundance. However, they contend this is being challenged by contemporary notions of affordability and sustainability which are replacing premium pricing and abundance as innovation’s drivers, particularly in emerging markets. As such they exhort firms to learn to do ‘more with less’ and for ‘more people’. They contend that "a potent combination of constraints and ambitions has ignited a new genre of innovation" (p.3).

A challenge to the assumptions of affluence and abundance means that the worldwide risks of a professional-use driven innovation (top-down) strategy for the existing global

multinational corporations have increased significantly. The huge research, development, prototype and global marketing costs, coupled with ever-increasing numbers of competing international players means that the amount of time during which a firm can enjoy its innovation rents is diminishing very rapidly (Fu et al, 2010). Initially, research and development and manufacturing moved abroad to tap talent or reduce costs but firms still focused mostly on products for wealthy nations. That is however changing in unique cases. Some organizations are experimenting by empowering R&D centres in emerging markets to take greater control and develop for home turfs while some local firms are innovating for local needs despite local limitations. An example of the former is General Electric's development of low-cost and portable ECG and ultrasound machines in India and China with eventual transfer to Western markets in what is called "reverse innovation" (Immelt, Govindarajan, & Trimble, 2009). An example of the latter is TATA's Nano car.

Van de Ven and Poole (1988) and Poole and Van de Ven (1989) believed that a paradox between different assumptions can be exploited towards theory development. Abrahamson (1991) used the paradox resolution approach and tools offered by Van de Ven and Poole (1988) and Poole and Van de Ven (1989) to come up with a theory of management fashion. We too take advantage of the contrasting assumptions on innovation to generate new theoretical perspectives for understanding innovation in emerging markets.

Poole and Van de Ven defined paradoxes as "interesting tensions, oppositions, and contradictions between theories which create conceptual difficulties" (1989:564). They suggested clarifying level of analysis, taking time into account, and introducing new terms as tools to stimulate theoretical development through paradox exploitation. Abrahamson (1991: 601) believes that this technique can "generate tailor-made theories for varied innovations or contexts". In contrast to the contingency approach of paradox resolution, this approach works

because it does not assume that different perspectives apply to some and not to other innovations or contexts.

At a very basic level our theoretical approach on juxtaposing the different perspectives of technology innovation, institutional innovation and social innovation helps us to not only understand each individually as they occur in emerging markets but, perhaps more importantly helps to develop an understanding of where "frugal innovation" may occur. We hypothesize that the 'ideal' frugal innovation depicted in case examples offered in mainstream practitioner literature lies at the intersection of all three, technology, social, and institutional innovations. Furthermore, our approach can help the study of diffusion of these innovations in varied contexts such as in production, service, profit, not-for-profit, or between developed and emerging market contexts.

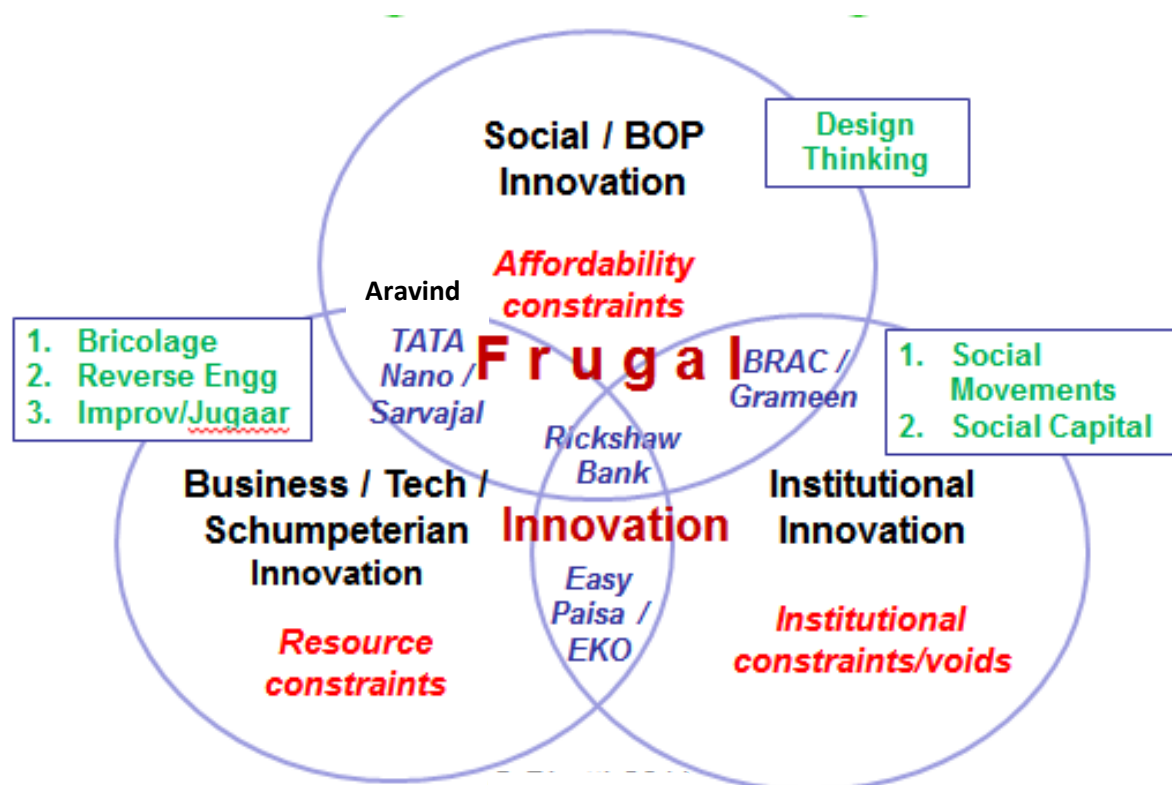
The Van de Ven and Poole approach to theorizing is powerful because it allows for "flexibility necessary to generate theories that incorporate assumptions that match the innovation or context they study and, thereby, provide for stronger empirical findings" Abrahamson (1991:601). As such we believe this theoretical approach we attempt here has strong potential for explanatory power particularly because the assumptions of resource constraints, institutional voids, and affordability concerns that underlie this theory match the context of emerging markets in which we hope researchers will test this theory.

The fact that so many organizations, both for-profit or non-profit, are looking to address the underserved (see for instance Collier, 2007; Easterly, 2006; Polak, 2008; Prahalad, 2005; Novogrotz, 2010), suggests we can learn from how technological, societal, and institutional fields are negotiating the space for frugal innovation. As such, we draw upon theories of resource-based view (Barney, 1991; Peteraf, 1993) resource-dependence (Pfeffer and Salancik, 1978), competitive advantage (Porter, 1995), institutional entrepreneurship

(DiMaggio, 1988) and institutional innovation (Hargrave and Van de Ven, 2006) to understand how firms may navigate resource, institutional, and social barriers or challenges.

Schumpeterian business helps to understand the resource constraints, institutional innovation the notion of institutional voids, and social innovation reveals the need and challenge of dealing with affordability constraints. However, we show how the three innovations taken separately cannot deal in solo with the challenges of innovating for the underserved in emerging markets. We analyse the intersections among these three innovation streams as fertile spaces where frugal innovation practises are occurring.

FIGURE 1: Theoretical model for Frugal Innovation



Intersection of Schumpeterian and Social Innovation

We begin with Schumpeter's (1934) seminal notion of innovation. This type is often associated with technology or business innovation (see for instance Christensen et al 2001; Geroski, 2003; and Utterback, 1994). Schumpeter attributed the innovative waves of "creative destruction" first to entrepreneurs and then to large corporations. The shift to large

corporations was in part because of the resources controlled by organizations. Yet today disruptive innovation is considered the playing ground for entrepreneurs. Regardless of who innovates, there is little debate on the need to access and control resources for innovation to occur. The procurement, control and combination of labor, skills, and material is crucial to the creation of new products and services (Schumpeter, 1934; Shane & Venkatraman, 2000).

Take for instance, entrepreneurs who need to mobilize resources to innovate and start new ventures (Shane and Venkatraman, 2000). Thus entrepreneurship researchers have analyzed how new ventures mobilize resources (Hsu, 2008; Shane, 2003) and the literature has focused on the mechanism of resource seeking, assuming often that resources are available in the environment. The probability of success in procuring these resources has been dependent on conditions such as the nature of the opportunity (Low and Abrahamson, 1997; Shane, 2000), venture legitimacy (Aldrich & Fiol, 1994; Meyer, 1983), founder experience (Amit, Glosten, & Muller, 1993; Boeker, 1989), and technological capability of the new venture (Shane & Stuart, 2002; Westhead & Storey, 1997).

Add to this the problem that many emerging and developing countries may not have the necessary resources available or the means to access any resources that might be available. An environment is resource constrained if it provides new challenges, whether opportunities or problems without providing additional or new resources (Baker and Nelson, 2005). In such contexts resource constraints are faced by both large corporations and entrepreneurs. One perspective is that given the relatively fewer resources available, emerging markets are developing their own less expensive technology and in the process decreasing reliance on imported costly technology (George and Prabhu, 2003). So despite resource-constrained environments and institutional challenges, ventures are creating affordable and innovative products and services. But how might they be doing so, given resource constraints?

Desa (2009) and Gundry et al (2011) show that even while lacking viable opportunities, legitimacy or intellectual property, many ventures stubbornly survive in penurious environments and are able to provide valuable products and services by relying on bricolage (Baker & Nelson, 2005, Levi-Strauss, 1967) as a resource mobilizing mechanism. Bricolage involves friends, family and volunteers for labour, purposely look for pre-existing materials for reuse, and build upon existing skills to allow a venture to originate and build capabilities for sustainability.

In extreme environments associated with developing countries, entrepreneurs who deploy bricolage are able to provide solutions that would be otherwise unavailable to users, for instance because of poverty or lack of access. This highlights the need to deal with both resource constraints as a process of innovation as well as affordability constraints as an outcome of innovation in emerging or developing markets. Desa writes, *"by rendering unique services in resource-poor environments, and by finding ways to maintain financial sustainability, entrepreneurs can create markets for services when previously none existed."* (p.201)

There is a growing consensus on addressing social objectives which improve living conditions. These social aspects can extend from societal needs such as justice and fairness to health, education and environmental and cultural preservation. Phillips et al (2008) define social value as the creation of benefits or reductions of costs for society in ways that go beyond the private gains and general benefits of market activity. If the benefits of these social objectives are to extend to society as a whole, they should also accrue to the disadvantaged or disenfranchised segments of society.

Social entrepreneurship meanings have ranged from the non-profit definition to the broader enactment of positive social change. Prahalad (2005) interprets social entrepreneurship to include entrepreneurship in emerging and developing markets that

improves the economic and social conditions of the poor and underserved communities. Concomitantly, social innovation encompasses new means and processes such as ideas, strategies, and organizations that enhance the social aspects of human function and civil society. It is defined in the Stanford Social Innovation Review as a "*novel solution to a social problem that is more effective, efficient, sustainable, or just than existing solutions and for which the value created accrues primarily to society as a whole rather than private individuals*" (Phills et al, 2008, p. 36). One example of social innovation is Ibrahim Abouleish's Sekem Initiative in Egypt. The community scheme reduces poverty, generates employability for rural farmers, and conserves natural resources. By applying organic agriculture techniques, Sekem contributed toward maintaining the regional natural environment (Seelos & Mair, 2005).

Social innovation embodies creative and innovative solutions to social problems, often operationalized through entrepreneurial activity. A financial return though important is often equally balanced or outpaced with the desire to achieve social impact. When social innovation serves the underserved, it often has to deal with the radical demand for lowering costs to meet local purchasing power. Although we do not discount the private gains through market activity, we believe that awareness of these opportunities to serve the underserved has attracted a number of efforts to address the problems of the poor through a combination of both market and non-market means (Mair and Marti, 2006). Combine social innovation concerns with technology innovation, and a space develops for products such as the Tata Nano and the Aravind Eye Hospital (see figure 1).

Phills et al differentiate social innovations from ordinary innovations because the world is already amply equipped to produce and disseminate ordinary innovations. They argue that when markets fail to produce for instance public goods, social innovation offers a way to meet needs and create value that would otherwise not be possible. Underserved and

neglected populations are unable to pay for basic goods and services such as healthcare, food and housing. Consequently, unfettered markets will not produce the goods and services for such populations. However with greater cooperation among non-profits, government, and business, new models of funding are triggering even profitable social innovations (Phills et al, 2008).

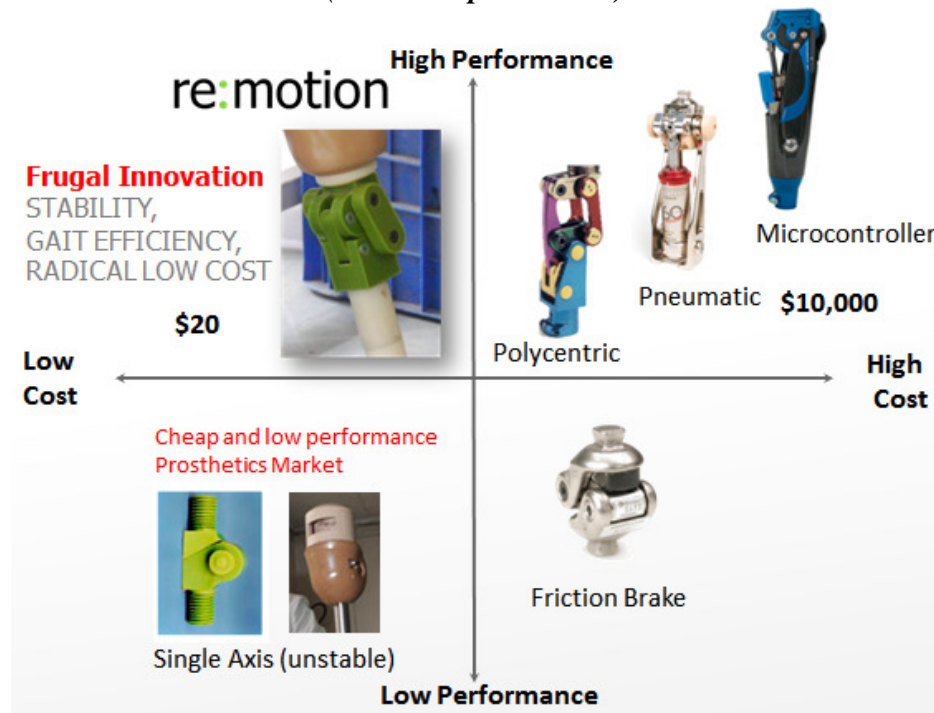
Why is that Aravind Eye Hospital is considered a social enterprise but not seen as a frugal innovation and the Tata Nano car or the One Laptop per Child is. *Perhaps because one is a consumer product and the other a social service?* Aravind reinvented the business model to serve upscale patients profitably and uses surplus revenue to subsidize free eye surgery for the bottom of the pyramid (Rubin, 2007; Mehta and Shenoy, 2011). But at the same time it reinvented the interocular implant to avoid an import cost of \$200 by building local capacity to manufacture at just \$5 a unit. We believe it was able to become a successful social enterprise by (frugally) innovating at the intersection of social and technological innovation.

We contend that economic and social changes have brought about a change of attitude in innovating at the intersections of sectoral divisions. Entrepreneurs and firms have begun to realize the benefit of innovating for such extreme customers, bottom of the pyramid, in extreme environments, emerging and developing markets. Working under these business and social constraints can force one to rethink both the process and outcome of innovation.

As Joel Sadler, designer of the \$20 Jaipur Remotion Knee and having worked at Apple, argues that working under formidable business constraints are what enables “extreme” products like the Apple Ipad, Airbook, and Iphone. Although Apple is more focused on performance metrics than price, it is there that Sadler learnt to innovate under the constraint of price-points as demanded by low-income customers. The result was a redesign of the prosthetic knee joint made of titanium with price ranges of \$5 thousand to a £20 low cost, simple, easily manufactured, and yet able to deliver high performance. It was noted as one of

the top 50 inventions by Time magazine 2009. See figure 2 of the frugal innovation space Jaipur Remotion knee positions itself, a space which was made obvious only from concerns emanating from their partner Jaipur Knee in India.

FIGURE 2: Price-performance space for Frugal Innovation ©Remotion
(used with permission)



Leaders of tomorrow should not be just innovators of products, services, technologies, but should also be innovators of institutions. Although institutions are the hardest to change, but that is where the greatest opportunities lie since those few who are able to change them, possess a capability that is very scarce (Haque, 2011). In researching over 250 companies, Haque found less than 20 that were institutional innovators. Such companies were not just marginally outperforming rivals, but structurally disrupting competition. In emerging and developing markets, institutional voids and market failures offer fertile ground and new spaces for business activity.

Intersection of Institutional and Social Innovation

Social enterprises may serve as the bridge between underserved communities and existing institutions (Wallace, 1999; Bayliss, 2004). Yet as ventures act to mobilize resources they brush against existing political (lobbying), legal (business regulation), and technological (human development) institutional environments, or the lack thereof, and in the process may build upon the existing fragments to create new structures. Further, several needs or problems in extreme environments are not addressed or even recognized by existing public or private institutions which leads firms having to deal and operate within this basic institutional void. Bricolage often goes against the norm and occurs in the absence of institutional support (Baker & Nelson, 2005; Baker, Miner, & Eesley, 2003).

Institutions that achieve consistency, impartiality, and reliability in their enforcement allow entrepreneurs to form expectations about the future, such as whether to invest in innovation, by removing some of the uncertainty about whether they will be able to capture the value they create (Baumol, 2002). Institutional entrepreneurship is one through which entrepreneurial actors change institutional structures or create new ones (DiMaggio, 1988; Fligstein, 1997). Institutional entrepreneurs “*lead efforts to identify political opportunities, frame issues and problems, and mobilize constituencies*” (Rao, Morrill, & Zald, 2000, p. 240). An institutional entrepreneur actor develops new institutions or facilitates change in existing institutions, and secures resources to achieve this change.

One stream of entrepreneurship literature looks at whether social entrepreneurship happens through or against existing institutions. While most ventures operate with the assumption their activities will be supported by formal institutions, other ventures may have to emerge despite a lack of institutional support (Sarasvathy, 2006). Research on institutional effects has shown that institutions can both support and preclude actors, entrepreneurs, and ventures. In such cases, institutional entrepreneurs emerge as a case of necessity.

Institutional entrepreneurs are actors who have an interest in modifying institutional structures and norms, or in creating new institutions, structure and norms (DiMaggio, 1988; Fligstein, 1997). When social entrepreneurship happens against existing institutional arrangements, the creation of a venture may in itself cause a change in that existing institutional arrangement (Mair and Marti, 2006) as occurred in the early days of Grameen (Yunus, 2006). Additionally, social entrepreneurship often takes place at the intersection of multiple institutions as it may have to deal simultaneously with the government, the market, and the community (Shaw and Carter, 2004). Mair et al (2011) call this having to deal with institutionally complex environments drawn from their work on BRAC in Bangladesh.

Sarasvathy (2006) points out that social enterprises are sometimes forced to go against existing institutions and come up with creative mechanisms that incorporate the best of both market and non-market solutions. Dean and McMullen (2007) show how according to the entrepreneurship literature, entrepreneurs can seize opportunities that are inherent in environmentally relevant market failures. Environmental degradation can be as the result of a market failure caused by existing institutional arrangements (Desa, 2009). As in the development of the recycling industry in the USA (Lounsbury, Ventresca, and Hirsch, 2003), social movements together with entrepreneurs can layout proofs-of-concepts for the creation of sustainable businesses. As this proof develops, wider public perception is changed, and consequently the institutions governing those perceptions are reconfigured to accommodate the new market.

Institutional analysis when bridged with social movement theory (e.g. Clemens, 1997; Rao, 1998; Strang and Soule, 1998; Armstrong, 2002) in sociology sheds an understanding of how political struggles shape cultural meaning systems, socio-economic processes, and industry emergence (Espeland and Stevens, 1998; Heimer, 2001; Lamont and Molnar, 2002).

The overlap of institutional and social innovation can be witnessed in historical accounts of the emergence of the recycling industry in the USA. Lounsbury, Ventresca, and Hirsch (2003) show how social movements have helped to transform socio-economic practices and paved the way for the recycling industry and its market development. The radical social movement for recycling promoted marginal practices through social-movement-inspired non-profit recyclers which then became central practices to the technology and by for-profit actors (Weinberg et al, 2000). In the emergence of the recycling industry labour intensive and often ad hoc non-profit voluntary recycling efforts were supplanted by more organized curb-side collection efforts promoted by for-profit models, yet depended on free household labour to clean and sort waste, thereby lowering the overall cost of the recycling process (Weinberg et al, 2000). The ability to leverage mainstream policy negotiation along with grassroots activism was key to the eventual success of the recycling movement (Lounsbury et al, 2003).

Microfinance pioneered by Muhammad Yunus and the Grameen Bank, both having shared the 2006 Nobel Peace Prize, we believe is one better explained as operating at the intersection of social and institutional innovations (see figure 1). The formation of a new microfinance institution and change of capital flows, lending, and savings patterns of the poor involves parallel efforts in institutional and social innovation.

Intersection of Schumpeterian and Institutional Innovation

Hargrave and Van de Ven (2006) juxtapose technology innovation literature and social movement literature to come up with a model for collective action model of institutional innovation. A difference in form, quality, or state over time in an institution, which is novel or an unprecedented departure from the past, is institutional innovation (2006).

Entrepreneurs and social activists both must engage in political processes that often resemble social movements (DiMaggio and Powell 1991; Fligstein 1996; 2001; Rao 1998;

Snow and Benford 1992). Both social movement and technology innovation management researchers have studied the institutional arrangements in which institutional change takes place, as well as the efforts of social activists and technological entrepreneurs to enact these arrangements. At this macro level of the inter-organizational field, the processes of technological innovation and entrepreneurship have many similarities to social movements (Aldrich and Fiol, 1994; Lounsbury and Ventresca, 2002; Schoonhoven & Romanelli, 2001).

Green (1992) has examined how a market may not exist for a new technology and may have to be shaped. For a market to exist, institutions must first be in place to establish prices, to inform customers and suppliers, and to provide distribution arrangements. Entrepreneurs developing radically new technologies must often engage in collective action with others to create these institutions. Bower and Christensen (1995) have examined the case of disruptive technologies, where information about potential customers, dimensions of product performance important to customers, and acceptable pricing level can only be acquired by experimenting with both the development of the product and the market.

The founder of IDEO, Tim Brown and head of IDEO's Social innovation group, Jocelyn Wyatt advise that when designing for developing nations, one not only consider form and function, but also distribution channels (Brown and White, 2010). They cite the example of northern Ghana where a programme for providing free nets to pregnant women and mothers with children under age 5 has been so successful that for everyone else, however, the nets are difficult to obtain. The reason is that for those not eligible to obtain free nets from public hospitals, there is no market for nets as it is not profitable to sell nets. Such is an example of where not considering the whole system, has had detrimental effects on those not within the scope of the current distribution scheme thereby making the eradication of malaria impossible (Brown and Wyatt, 2010).

In the US, the emergence of new internet business models brought about by Ebay for everyday selling and E-Trade for investments options to those often excluded from the markets is made possible by innovating at the intersection of technological and institutional arrangements (figure 1). In emerging markets examples of mobile phone banking, M-Pesa in Kenya, Telenor EasyPaisa in Pakistan and EKO in India, though all three are technology companies, innovate at both the frontiers of technology and institutions. They helped connect the poor “unbanked” sector to existing financial institutions, or lack thereof, by leveraging the ubiquity of mobile technology.

Telenor EasyPaisa worked with the government to devise the world’s first regulatory mobile banking model. The company mobilized more than 25,000 (and growing) of its franchises in just two years to serve as “bank kiosks” in a country where only 4,000 traditional bank branches serve 180 million people. Easypaisa have brought in more people into the formal economy than ever possible through traditional institutionalized bank models alone. EKO has bypassed the mobile companies to create a mobile phone app that works on any network (and any phone) that links major banks such as ICICI with thousands of growing mom and pop shop owners who serve as bank tellers. These frugal innovations created different business models in each of their respective environments ranging from the absence of bank involvement in Kenya, mobile led in Pakistan, and entrepreneurial broker led in India. Yet all help lower entry barriers for the financially underserved to benefit from savings, investments, and insurance markets for more inclusive growth. EKO’s CEO Abhishek Sinha claims their most socially valuable customers are household domestic maids who can now save money in their own bank accounts and access their funds virtually throughout India.

Eric Topol (2011) Chief Academic Officer of Scripps Health and Professor of Genomics cites a \$8K pocket ultrasound device as an example of frugal innovation that does

just a decent job of visual scans as a single \$2K professional and technical scan done by conventional machines in a US laboratory. He believes although such "frugal innovations" abound in the healthcare sector, their use or adoption is hindered by the reimbursement rules and economic models that physicians, insurers, and hospitals rely upon. Therefore although these devices are being readily developed and adopted in emerging markets, in the West eventual adoption depends not only on technological prowess of frugal innovations but also in restructuring or working around existing institutional structures.

The Social-Schumpeterian-Institutional Nexus

Although isolation frees an emerging system from the institutional constraints of existing technologies and industries (Astley, 1985) and permits it to develop its own distinctive institutional forms (Rappa, 1987), we believe it is best considered in combination with social and business concerns.

We contend that social innovations lie at one extreme of the spectrum where social value as a whole is of the greatest concern while ordinary innovations lie on the other extreme where profits and value accrue to individuals, often privileged ones who already are advantageously within the folds of markets. In the middle of this lies frugal innovation, which draws social value through a mix of social, business and institutional innovation. Innovation blossoms at the intersections where the sectors converge because of exchange of ideas and values, shifts in roles and relationships, and the integration of private capital and public and philanthropic support (Phills et al, 2008).

McMullen (2011) offers a theory of development entrepreneurship which lies at the intersection of social, institutional, and business entrepreneurship. Similarly, frugal innovation resides at the intersection of social entrepreneurship, technology entrepreneurship, and institutional entrepreneurship, all emerging fields in themselves, but together offering challenging opportunities for research to test and refine theories of entrepreneurship,

innovation, and strategy in emerging market contexts. Mair, Hockerts & Robinson (2006) and Shane & Venkatraman (2003) pose the challenge for researchers to apply theories of management to situations that are relevant to social entrepreneurs and technology entrepreneurs. But add to this the fact that social enterprises routinely lie at the nexus of the public and private sectors (Dees, 1998).

We therefore pose a further challenge to coincide social, technology, and institutional innovations. In technological innovation research it is proposed that knowledge bridging as a boundary-spanning mechanism can produce radical innovations and creative products (Hargadon & Sutton, 1997; Hsu & Lim, 2005). As emerging and developing country pro-poor ventures operate at the boundaries of resource, affordability, and institutional constraints, we propose that knowledge-bridging mechanisms across the three types of innovation, social, technical, and institutional help to explain the development of frugal innovations, particularly in emerging markets, but possibly not limited to.

In the mid-1990s in the US, Self-Help, a community development finance organization pioneered the secondary market for mortgage-backed securities based on loans to low-income households to enable greater access to homeownership. With cross sector partnerships between the non-profit Self-Help, commercial banks, federally chartered Fannie Mae, and private investors, the scheme grew from \$50m in 1998 to \$2.5b in 2008 (Phills et al, 2008). Such a sophisticated scheme that relies on robust and varied institutions would unlikely work in emerging and developing markets. In contrast however, through a combination of technical, social and community based social innovations, for instance, low income housing schemes are being experimented with in emerging markets. Bahria Town of Pakistan, the largest private real estate developer in Asia, has packaged together technical innovations to lower the cost of housing construction, social innovations to stymie the influx

of real estate speculators, and institutional innovation to enable financing for low-income household ownership.

Another example is the Rickshaw Bank project in India which we believe exemplifies a position at this nexus of social, technical, and institutional innovations (figure 1). Sarmah (2010) set out to enable rickshaw drivers, most of whom are immigrant workers from the rural regions, to own rather than rent rickshaws. In the process, he used technology to redesign the rickshaw, social innovation to restructure the prevalent practise of renting to owning, and used institutional innovation to license and recognize rickshaw owners as formal contributors to the local economy. The rickshaw community has thus been drawn from a bleakly structured informal economy and into the folds of the formal economy. This has also led the rickshaw owners to rent clean energy LPG cylinders at subsidized government prices instead of using firewood or expensive kerosene. Sarmah's rickshaw bank enterprise provides not only rent to buy rickshaws, but also damage and healthcare insurance. Any of the three innovations in isolation would have not likely have succeeded yet by combining the three, Sarmah has been able to help more than 30,000 rickshaw drivers to benefit from the scheme. Plans are underway to use the micro-franchise model and scale to other parts of India to reach the remaining 8 million or so rickshaw drivers.

We defined frugal innovation as one that redefines business models, reconfigures value chains and redesigns products to use resources in different ways and create more inclusive markets by serving users with affordability constraints, often in a scalable & sustainable manner. The Rickshaw Bank project has done just that, redefine business model, reconfigure value chains and redesign products to create more inclusive markets. Scalability, although the model seems scalable, we need to see to what extent Sarmah's business model will diffuse to the majority of market clients.

IMPLICATIONS AND FUTURE RESEARCH

This paper may contribute to insights into theory building in innovation and to understanding management issues (Van de Ven, 1986; Leonard-Barton, 1988) as applied on emerging market contexts. We applied Poole and Van de Ven's (1989) logic of paradox resolution to develop a theory of frugal innovation which better captures the complexity of different assumptions involved in understanding the phenomenon of innovation in emerging markets.

In terms of theory, there is potential to demonstrate that a certain kind of innovation process is operating in emerging markets -- frugal innovation -- the impact of which may diffuse to developed markets. In contrast to the view that innovation occurs only when slack resources and supporting institutions exist, and that high income users are often the first target segment of choice, frugal innovation shows all three are not necessary to producing globally useful solutions. In contrast to innovating only when the right components of a National Innovation System (Lundvall, 1992; Freeman, 1995) are present, the model herein enables recognition of an approach to innovation which is catching on in particularly Asia despite and because of extreme social needs, resource constraints, and institutional voids.

In understanding what the space for frugal innovations looks like and how it can be organized, we proposed a model to theoretically conceptualize 'frugal innovation'. The model attempts to apply (and in later work test) theories in emerging markets to try and understand innovation trends unique to those contexts. The examples we presented as fitting in the different intersections of the typological model are based on exploratory research and therefore tentative. However, the exercise does show the overlapping spaces within which frugal innovation activities are carried out.

We drew upon strategy (resource constraints), entrepreneurship (social needs), and institutional (voids) literatures to propose a theoretical framework to analyse innovation in the extreme or penurious contexts of emerging or developing markets. We conceptualize the nexus of social, institutional, and business innovation as fertile overlaps that aid in

identifying, understanding and positioning frugal innovations. Empirical studies that test this theory and its ability to generalize beyond emerging market contexts are called for. Research can inform on the success and failures of frugal innovation strategies and of the necessary capabilities needed for firms to adopt them.

In terms of practice, the model we offer to understand frugal markets and innovation can help firms and entrepreneurs seeking to innovate in emerging markets, analyze and categorize the different approaches to innovation based on which nexus of constraints they need to address and leverage. Should contextual factors and demands change or should a firm wish to enter different or home markets, an awareness of the need to reposition to other subsets can be valuable for strategy reformulation.

As the West looks to the East for innovation, ideas, and entrepreneurship (Capelli et al, 2010; Ming-Jer and Miller, 2010), frugal innovation could offer fresh ideas and new perspectives in cost minimization and innovation maximization to a world slowed down by the recent recession, i.e. doing more with less. In some ways, the activity of frugal innovation is already catching on in the West in its own way. The current trend of collaborative consumption where car (RelayRides), house (Airbnb), and baby clothes sharing (Plumgear) are brokered by web start-ups in San Francisco who use existing publicly owned assets are in many forms and practices frugal innovation.

Among the most influential social scientists is Daniel Kahneman who won the 2002 Nobel laureate for "having integrated insights from psychological research into economic science" (Goldstein, 2011). According to the Thomson Reuters Web of Science, Kahneman and his prospect theory has been cited in scholarly journals more than 28,312 times since 1979. Jevin West of the University of Washington helped develop an algorithm for tracing the spread of ideas among disciplines and comments on Kahneman, "Kahneman's career shows that intellectual influence is the ability to dissolve disciplinary boundaries" (Goldstein,

2011). In much the same way, we believe our attempt to understand innovation in emerging markets through the proposed theoretical model dissolves the disciplinary boundaries of technology, social, and institutional innovation literatures.

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